

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.C.A. (CBCS) Degree Examinations
 January 2024

22MCA104 – MATHEMATICAL FOUNDATION FOR COMPUTER APPLICATIONS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Use set builder notation and logical equivalence to establish the first De Morgan law $\overline{A \cap B} = \overline{A} \cup \overline{B}$	05	L*1	1	1
b) Find the Boolean product of A and B, where				
$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$	5	L2	1	1
c) Find eigen value and eigen vector of the following 2*2 matrix.	10	L2	1	1
$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$				
2. a) A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Russian, and 14 have taken courses in both French and Russian. If 2092 students have taken at least one of Spanish, French, and Russian, how many students have taken a course in all three languages?	10	L2	1	1
b) What is the Cartesian product $A \times B \times C$, where $A = \{0,1\}$, $B = \{1,2\}$, and $C = \{0,1,2\}$?	05	L2	1	1
c) With example explain computer representation of sets. We have seen that the bit string for the set $\{1,3,5,7,9\}$ (with universal set $\{1,2,3,4,5,6,7,8,9,10\}$) is 10 1010 1010.				
What is the bit string for the complement of this set?	05	L2	1	1

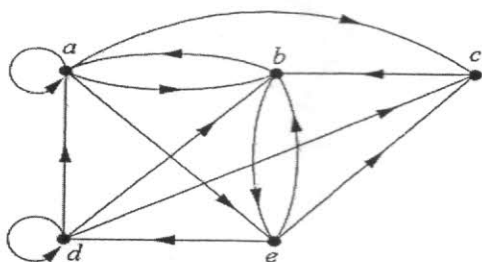
Unit – II

3. a) Translate the following statement into English, where the domain for both variables consists of all real numbers. $\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0)).$	05	L1	2	1
b) What are the contrapositive, the converse, and the inverse of the conditional statement "The home team wins whenever it is raining?"	05	L1	2	1
c) Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.	10	L2	2	1
4. a) Check whether the logical expressions $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent by drawing the truth table.	10	L1	2	1
b) Build a digital circuit that produces the output $(p \vee r) \wedge (\neg p \vee (q \vee r))$ when given input bits p, q, and r.	10	L2	2	1

Unit – III

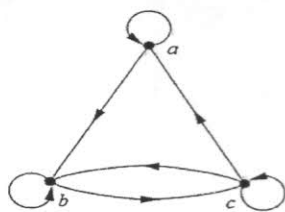
5. a) Let A be the set $\{1,2,3,4\}$. Which ordered pairs are in the relation $R = \{(a,b) \mid a \text{ divides } b\}$? Display the result in graphical and tabular form.	10	L1	3	1
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- 5) Which of the following are paths in the directed graph shown below : a,b,e,d; a,e,c,d,b; b,a,c,b,a,a,b; d,c; c,b,a; e,b,a,b,a,b,e? What are the lengths of those that are paths? Which of the paths in this list are circuits?

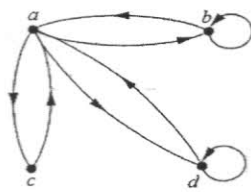


10 L2 3 1

6. a) Determine whether the relations for the directed graphs shown below are reflexive, symmetric, antisymmetric, and/or transitive.



(a) Directed graph of R



(b) Directed graph of S

10 L2 3 1

- b) Find the zero-one matrix of the transitive closure of the relation R where

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

10 L1 3 1

Unit – IV

7. a) Find mean and variance of the discrete random variable X.

X	1	2	3	4	5	6	7	8	9	10
P(x)	.15	.1	.1	.01	.08	.01	.05	.02	.28	.2

10 L2 4 1

- b) If X is a random variable on a sample space S, then $V(X) = E(X^2) - E(X)^2$.

05 L1 4 1

- c) A fair coin is flipped three times. Let S be the sample space of the eight possible outcomes, and let X be the random variable that assigns to an outcome the number of heads in this outcome. What is the expected value of X?

05 L1 4 1

8. a) State and prove Bayes theorem.

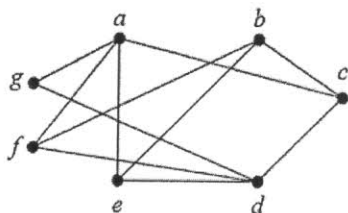
10 L2 4 1

- b) What is the expected value of the sum of the numbers that appear when a pair of fair dice are rolled?

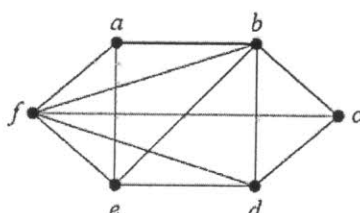
10 L2 4 1

Unit – V

9. a) Whether the graphs G and H shown below are bipartite? Give reasons.



G



H

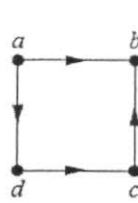
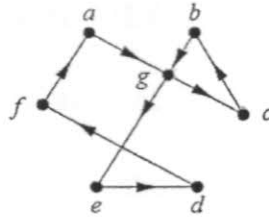
10 L1 5 1

- b) Explain each with examples

- Bipartite graph
- Adjacency matrices

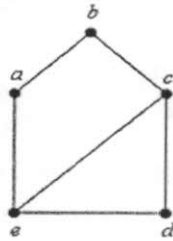
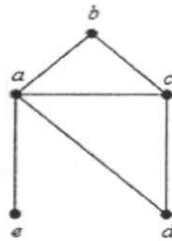
10 L2 5 1

- a) Define Euler circuit and Euler path. Which of the directed graphs shown below have an Euler circuit? Which of them have an Euler path?

 H_1  H_2  H_3

10 L2 5 1

- b) Justify whether the graphs displayed below are isomorphic.

 G  H

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
